SylvanSync™

CONTINGENCY LESSONS

Mathematics for SylvanSync™

Intermediate

Sylvan Learning[sm]

GRADES 6-8

To be used only in case of emergency, with the approval of Technology Services.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Operations with Decimals and Fractions

Objectives

The student will be able to:

- add, subtract, multiply, and divide mixed numbers.
- add and subtract decimals through thousandths.

Materials

- *VersaTiles: Operations with Fractions* book (VT6OF)
- *VersaTiles: Operations with Decimals* book (VT6OD)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself! Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn how to add, subtract, multiply, and divide with mixed numbers and decimals.

Add:

$3\frac{1}{2}+4\frac{2}{3}$

Write equivalent fractions with like denominators, then add. Regroup improper fractions in the sum: $3\frac{1\times 3}{2\times 3}+4\frac{2\times 2}{3\times 2}=3\frac{3}{6}+4\frac{4}{6}=7\frac{7}{6}=8\frac{1}{6}$

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Operations with Decimals and Fractions

Multiply:

$2\frac{1}{5}\times 1\frac{3}{8}$

Write mixed numbers as improper fractions, then multiply numerators and denominators. Regroup improper fractions in the product:

$\frac{(2\times 5)+1}{5}\times\frac{(1\times 8)+1}{8}=\frac{11}{5}\times\frac{9}{8}=\frac{99}{40}=2\frac{19}{40}$

Subtract:

5.365 – 2.24

Align the decimal points, then subtract place by place:

$$\begin{array}{r} 5.365 \\ -2.24 \\ \hline 3.12 \end{array}$$

Activity

Go to page 11 in *VersaTiles: Operations with Fractions*. Match the addition problem to the sum. Flip over the box to see if you did it correctly.

Go to page 14 in *VersaTiles: Operations with Fractions*. Match the subtraction problem to the difference. Flip over the box to see if you did it correctly.

Go to page 22 in *VersaTiles: Operations with Fractions*. Match the multiplication problem to the product. Flip over the box to see if you did it correctly.

Go to page 31 in *VersaTiles: Operations with Fractions*. Match the division problem to the quotient. Flip over the box to see if you did it correctly.

Go to page 5 in *VersaTiles: Operations with Decimals*. Match the addition problem to the sum. Flip over the box to see if you did it correctly.

Go to page 8 in *VersaTiles: Operations with Decimals*. Match the subtraction problem to the difference. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Data and Chance

Objectives
The student will be able to:

- read and interpret data displays.
- find measures of center for a set of data.

Materials
- *VersaTiles: Data and Chance* book (VT6DC)
- VersaTiles

Background Information
VersaTiles let you learn, practice, and review skills all by yourself! Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn how to read and interpret data displays and find measures of center for a set of data.

Find the mean, median, and mode of the set of data.

4, 4, 5, 7, 8, 8, 8, 9, 10, 12

Mean is another word for average. To find the mean add all the numbers in the set and divide by the number of elements in the set.

$$\frac{4+4+5+7+8+8+8+9+10+12}{10} = \frac{75}{10} = 7.5$$

Median is the value at the middle of an ordered data set. To find the median list the elements of the set in order from least to greatest. Find the element in the middle of the data set. The median of our data set is 8.

Mode is the element that occurs the most frequently in the data set. To find the mode, list the elements of the data set in order from least to greatest and find the element that appears the most often. The mode of our data set is 8.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Data and Chance

Activity

Go to page 5 in *VersaTiles: Data and Chance*. Match the question about the data in the double-bar graph to the answer. Flip over the box to see if you did it correctly.

Go to page 8 in *VersaTiles: Data and Chance*. Match the question about the data in the double-line graph to the answer. Flip over the box to see if you did it correctly.

Go to page 9 in *VersaTiles: Data and Chance*. Match the question about the data in the circle graph to the answer. Flip over the box to see if you did it correctly.

Go to page 13 in *VersaTiles: Data and Chance*. Match the data set to the mode. Flip over the box to see if you did it correctly.

Go to page 14 in *VersaTiles: Data and Chance*. Match the data set to the mean. Flip over the box to see if you did it correctly.

Go to page 17 in *VersaTiles: Data and Chance*. Match the data set to the median. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Integer Operations

Objectives
The student will be able to:

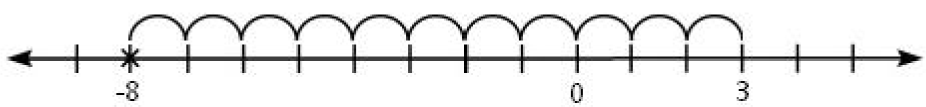
- add and subtract integers using a number-line model.
- multiply and divide integers using a counter model and rules of integer operations.

Materials
- *VersaTiles: Integer Operations* book (VT7IO)
- VersaTiles

Background Information
VersaTiles let you learn, practice, and review skills all by yourself! Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn how to perform operations with integers.

Add 3 + (–11).

Start at 3. Move 11 places to the left. 3 + (–11) = –8.



Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Integer Operations

Activity

Go to page 5 in *VersaTiles: Integer Operations*. Match the addition problem to the sum. Flip over the box to see if you did it correctly.

Go to page 13 in *VersaTiles: Integer Operations*. Match the subtraction problem to the answer. Flip over the box to see if you did it correctly.

Go to page 21 in *VersaTiles: Integer Operations*. Match the multiplication problem to the product. Flip over the box to see if you did it correctly.

Go to page 22 in *VersaTiles: Integer Operations*. Match the multiplication problem to the product. Flip over the box to see if you did it correctly.

Go to page 27 in *VersaTiles: Integer Operations*. Match the division problem to the quotient. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Ratios, Proportions, and Percents

Objectives

The student will be able to:

- calculate and compare unit rates.
- write a proportion and use a proportion to find an unknown value.
- calculate a percent of a number.

Materials

- *VersaTiles: Ratios, Proportions, and Percents* book (VT7RPP)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself! Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn how to calculate and compare units rates; to write a proportion and use a proportion to find an unknown value; and to calculate a percent of a number.

Compare Unit Rates

Which is a better buy, four pounds for $9.00 or three pounds for $7.50?

Calculate the unit rates:

$$\frac{\$9.00}{4\text{ lb}} = \frac{\$9.00 \div 4}{4\text{lb} \div 4} = \frac{\$2.25}{1\text{ lb}} \qquad \frac{\$7.50}{3\text{ lb}} = \frac{\$7.50 \div 3}{3\text{ lb} \div 3} = \frac{\$2.50}{1\text{ lb}}$$

Compare the unit rates. The lower unit rate is the better buy. Four pounds for $9.00 is the better buy.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Ratios, Proportions, and Percents

Solve a Proportion

What is the value of *x*?

$$\frac{x}{12} = \frac{4}{3}$$

Use the cross-products.

$$\begin{aligned} 3x &= 12 \times 4 \\ 3x &= 48 \\ 3x \div 3 &= 48 \div 3 \\ x &= 16 \end{aligned}$$

Calculate a Percentage

What is 20% of 45?

20% of 45 = *n*

0.2 × 45 = *n*

9 = *n*

Activity

Go to page 8 in *VersaTiles: Ratios, Proportions, and Percents*. Match the pair of rates to the lower or higher unit rate. Flip over the box to see if you did it correctly.

Go to page 11 in *VersaTiles: Ratios, Proportions, and Percents*. Match the proportion to the ratio that completes it. Flip over the box to see if you did it correctly.

Go to page 13 in *VersaTiles: Ratios, Proportions, and Percents*. Match the proportion to the missing term. Flip over the box to see if you did it correctly.

Go to page 24 in *VersaTiles: Ratios, Proportions, and Percents*. Match the numbers to the percent. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Algebra and Functions

Objectives

The student will be able to:

- write an equation to represent a situation.
- solve one-step equations.
- solve two-step equations.

Materials

- *VersaTiles: Algebra and Functions* book (VT8AF)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself! Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn how to write and solve equations.

Write an equation to represent the statement, “Four more than twice a number is 16.”

$4 + 2x = 16$

Solve the equation:

$$\begin{aligned} j + 4.5 &= 11.6 \\ j + 4.5 - 4.5 &= 116 - 4.5 \\ j &= 7.1 \end{aligned}$$

Mathematics for SylvanSync™
Grades 6–8
VersaTiles: Algebra and Functions

Activity

Go to page 5 in *VersaTiles: Algebra and Functions*. Match the statement to the equation. Flip over the box to see if you did it correctly.

Go to page 6 in *VersaTiles: Algebra and Functions*. Match the equation to the solution. Flip over the box to see if you did it correctly.

Go to page 8 in *VersaTiles: Algebra and Functions*. Match the equation to the solution. Flip over the box to see if you did it correctly.

Go to page 14 in *VersaTiles: Algebra and Functions*. Match the equation to the solution. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 6–8
The Problem-Solving Process

Objectives
The student will be able to:

- identify the four basic steps of the problem-solving process.
- use the problem-solving process to solve problems.

Materials
- none

Background Information
There are four basic steps to solving word problems.

- **Step 1:** Understand the problem.
  Identify what you know and what you need to know.

- **Step 2:** Develop a plan.
  Choose a strategy and estimate the answer.

- **Step 3:** Solve the problem.
  Apply the strategy and find a solution.

- **Step 4:** Look back.
  Make sure that you have answered the question asked and that your answer is reasonable.

Mathematics for SylvanSync™
Grades 6–8
The Problem-Solving Process

Activity
Complete the activities that follow on a separate sheet of paper.

Review and use the problem-solving process to solve problems.

Part 1
Review the four steps to solving any problem. Write the number of the step in which you would do each of the following.

_____ Perform calculations.

_____ Choose a strategy.

_____ Check your work.

_____ Find the information given.

Part 2
Brielle bought a sandwich for $4.25, a drink for $1.50, and fruit salad for $2.75. She gave the clerk $10.00. How much change did she receive?

Step 1: Understand the problem.

1. a. What information do you have? Write the facts.

   b. What are you trying to find?

Step 2: Develop a plan.

2. How will you solve the problem?

Step 3: Solve the problem.

3. How much change did Brielle receive?

Step 4: Look back.

4. Is your answer reasonable? Why or why not?

Mathematics for SylvanSync™
Grades 6–8
The Problem-Solving Process

Part 3

Solve each problem. Show all four steps of the problem-solving process and write your answer in a complete sentence.

1. Tyler ran for 30 minutes on Monday, 40 minutes on Tuesday, 20 minutes on Wednesday, 25 minutes on Thursday, and 50 minutes on Friday. What was the average number of minutes he ran each day?

2. Hannah wants to buy a sweater that costs $25 and a pair of pants that costs $20. She earns $8 per hour babysitting. For how many hours will she need to babysit to earn enough money to buy the sweater and pants?

3. Irina is going to visit her grandmother who lives 240 miles away. She travels 100 miles on the first day and 86 miles on the second day. How many more miles does she need to travel?

4. A square sheet has a side length of 8 feet. A square-shaped section with a side length of 3 feet is cut out of the center of the sheet. What area of the sheet remains?

Mathematics for SylvanSync™
Grades 6–8
Problem-Solving Strategies

Objectives
The student will be able to:

- choose an appropriate problem-solving strategy.
- use problem-solving strategies to solve word problems.

Materials
- none

Background Information
The second step in the four-step problem-solving plan is to develop a plan. Part of developing a plan is choosing an appropriate problem-solving strategy. Problem-solving strategies include Make a Model, Guess and Test, Write an Equation, Make a List, Work Backward, Solve a Simpler Problem, Logical Thinking, Draw a Diagram, Find a Pattern, Make a Table, Simulation, and Draw a Picture.

Mathematics for SylvanSync™
Grades 6–8
Problem-Solving Strategies

Activity
Complete the activities that follow on a separate sheet of paper.

Choose an appropriate problem-solving strategy for a given problem and apply the strategy to solve the problem.

Part 1
Review the problem-solving process. Write the name of each step on your paper.

Step 1:

Step 2:

Step 3:

Step 4:

Part 2
Jennifer goes swimming every three days. She rides her bike every four days. How many times in 30 days will she do both activities?

1. Step 1:
   a. What information are you given?
   b. What do you need to find?
2. Step 2: Choose an appropriate problem-solving strategy.
3. Step 3: Solve the problem using the strategy you chose in Step 2. Write your answer in a complete sentence.
4. Step 4: Is your answer reasonable?

Mathematics for SylvanSync™
Grades 6–8
Problem-Solving Strategies

Part 3

For each problem, choose an appropriate problem-solving strategy, and then solve. Write the name of the strategy you plan to use, show your work, and write your answer in a complete sentence.

1. Mrs. Ellison has a bathroom that is 60 inches wide and 90 inches long. She is tiling the bathroom floor with tiles that are 12 inches wide and 15 inches long. How many tiles will Mrs. Ellison need to cover her floor with no gaps and no overlap?

2. Mitchell is packing for a vacation. He is packing four pairs of pants, six shirts, and two sweaters. How many different outfits will Mitchell be able to make if he chooses one pair of pants, one shirt, and one sweater for each outfit?

3. Alyssa had some bracelets. She gave half of them to her sister. Then she bought three more. After giving five bracelets to her cousin, she had seven left. How many bracelets did Alyssa start with?

4. Milo can do eight math problems in an hour. Working at this rate, how long will it take him to complete a homework assignment with 24 problems?

Mathematics for SylvanSync™
Grades 6–8
Pythagorean Theorem

Objectives
The student will be able to:

- represent the Pythagorean Theorem using a model.

Materials
- Pythagorean Theorem model
- grid paper
- scissors

Background Information
A triangle is a two-dimensional figure that has three sides and three angles. Triangles can be classified by their angles. If a triangle has three acute angles, then it is an acute triangle. If a triangle has one obtuse angle then it is an obtuse triangle. If a triangle has one right angle then it is a right triangle. The sides of a right triangle are named according to the angles they are opposite. The side opposite the right angle is called the hypotenuse. The hypotenuse is the longest side of the triangle. The other two sides, opposite the two acute angles, are called legs.

Mathematics for SylvanSync™
Grades 6–8
Pythagorean Theorem

Activity
Complete the activities that follow on a separate sheet of paper.

Part 1
Look at the Pythagorean Theorem model.

1. Which side is the hypotenuse? Which sides are the legs? How do you know? Write your answer in complete sentences.
2. Each square in the model has a side length of one unit. How long is the hypotenuse? How long are the legs?
3. Count the number of smaller squares in each of the three larger squares. Record the number of squares adjacent to the hypotenuse and each of the legs.
4. How are the numbers of smaller square related? Write a true equation using the three numbers.
5. Write a sentence that explains the relationship among the three numbers.
6. Write the number of squares at each side as an expression using an exponent. Then, write an equation relating the three expressions. (Hint: Look at your equation from Question 4.)

Mathematics for SylvanSync™
Grades 6–8
Pythagorean Theorem

Part 2

Draw a Pythagorean Theorem model.

1. On the grid paper, draw the legs of a right triangle. Each line should be drawn along a grid line. Draw one leg that is 5 squares long and one leg that is 12 squares long. Connect the ends of the legs to draw the hypotenuse.
2. Draw a square on each leg of the triangle. Cut a square to place on the hypotenuse of the triangle.
3. Count the number of grid squares in each larger square. Record your answers.
4. Write an equation using the numbers of smaller squares.
5. Write the equation again, this time using exponential expressions for the numbers of squares.
6. Is the relationship the same as the relationship you found in Part 1? Write your answer in a complete sentence.

Mathematics for SylvanSync™
Grades 6–8
Pascal's Triangle

Objectives
The student will be able to:

- identify patterns in Pascal’s triangle.
- make predictions using the numbers in Pascal’s triangles.

Materials
- crayons or markers
- copy of Pascal’s Triangle Worksheet

Background Information
Pascal’s triangle is named after a seventeenth century French mathematician, Blaise Pascal. Its numbers are arranged in rows. Each row begins and ends with the number one and every other number is the sum of the two numbers above it.

Mathematics for SylvanSync™
Grades 6–8
Pascal's Triangle

Activity
Complete the activities that follow on a separate sheet of paper.

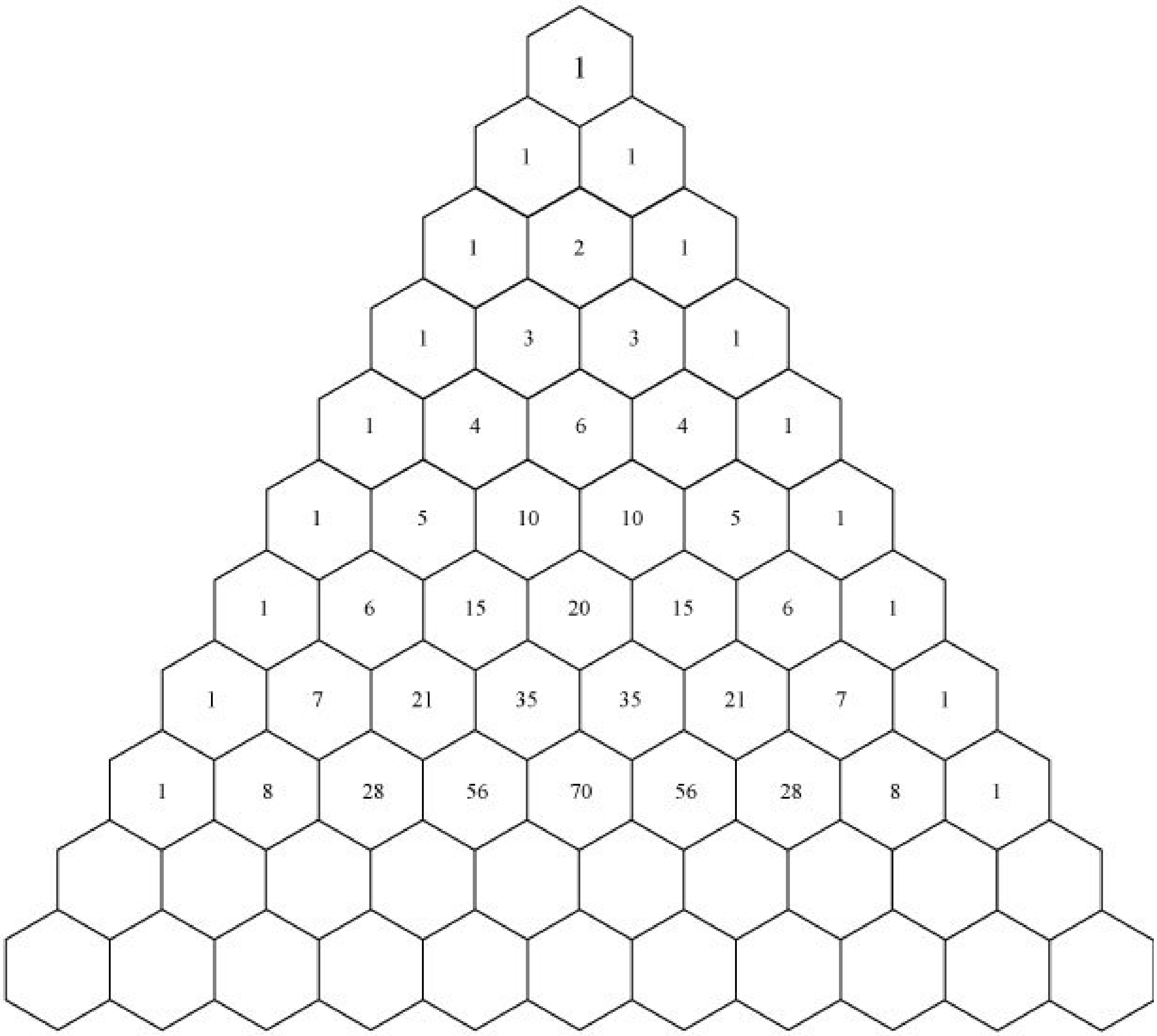
Explore the patterns in Pascal’s triangle.

Part 1
1. Complete the next two rows in the triangle on the worksheet.
2. What did you notice about the first five numbers in each of the rows you just completed, compared to the last five numbers in those rows?
3. The outside numbers along the left side of the triangle are the first diagonal. What do you notice about the numbers in the second diagonal?
4. What pattern do you notice in the numbers in the third diagonal?
5. The numbers in the third diagonal are called triangular numbers. Why do you think this is so?
6. Find the sum of each of the first six rows in the triangle. What do you notice?
7. Make a prediction about the sum of the numbers in the seventh row. Find the sum to check your prediction.
8. Lay your pencil over the vertical line of symmetry in the triangle. What do you notice about the numbers on either side of the pencil?
9. Select your favorite color and shade the multiples of three. What pattern do you notice?

Optional Activity
1. Select a second color and shade numbers that are one less than the multiples of three.
2. Select a third color and shade numbers that are two less than the multiples three.
3. What patterns do you notice in the design?

Mathematics for SylvanSync™
Grades 6–8
Pascal's Triangle

Pascal's Triangle Worksheet



Mathematics for SylvanSync™
Grades 6–8
Exploring Integers

Objectives
The student will be able to:

- represent positive and negative integers with two-color counters.
- add and subtract integers using two-color counters.

Materials
- two-color counters

Background Information
Integers are the counting numbers and their opposites. Integers that are greater than zero are positive, and numbers that are less than zero are negative. The absolute value of an integer is its distance from zero. Integers with the same absolute value and different signs are called “opposites.” To add integers, represent each integer and then remove pairs of red and yellow counters until only one color remains. The remaining counters represent the sum. To subtract an integer, add its opposite.

Mathematics for SylvanSync™
Grades 6–8
Exploring Integers

Activity
Complete the activities that follow on a separate sheet of paper.

Use two-color counters to represent integers and to add and subtract integers. Each yellow counter has a value of +1, and each red counter has a value of –1.

Part 1
1. Use counters to represent each integer below.

   a. +6 b. –4 c. 11 d. –9

2. Write the integer that each group of counters represents.

   a. 2 red counters b. 5 yellow counters c. 10 red counters

Part 2
1. Use counters to model each pair of integers. Find the sum. Then, write an equation to represent the addition. Remember that one yellow counter paired with one red counter has a zero sum.

   a. 4, –10 b. 3, 6

   c. –7, –5 d. –2, 9

2. What do you notice about the sign of the sum in each case?

Mathematics for SylvanSync™
Grades 6–8
Exploring Integers

Part 3

1. Use counters to model each subtraction. Find the difference. Then, write an addition equation to represent the subtraction. Remember that you may need to add zero sum pairs of counters before you can subtract.

   a. 4 – (+7)

   b. 7 – (+4)

   c. 12 – (–3)

   d. –3 – 12

   e. –8 – (+10)

   f. 10 – (–8)

   g. –6 – (–5)

   h. –5 – (–6)

2. What do you notice about the sign of the difference if the order of the integers is reversed?

Mathematics for SylvanSync™
Grades 6–8
Fractions with Number Cubes

Objectives
The student will be able to:

- add and subtract fractions with like denominators.
- add and subtract fractions with unlike denominators.
- compare fractions.
- multiply and divide fractions.

Materials
- a pair of dice

Background Information
This lesson involves operations with fractions. To add or subtract fractions with like denominators, perform the operation on only the numerators of the fractions. To add or subtract fractions with unlike denominators, first find a common denominator, and then write equivalent fractions with the common denominator before performing the operation on the new numerators. To multiply fractions, multiply the numerators of the two fractions and the denominators of the two fractions. To divide fractions, multiply the first fraction by the reciprocal of the second fraction. Express all answers in simplest form.

Activity
Complete the activities that follow on a separate sheet of paper.

Use two dice to generate fractions, and then add, subtract, multiply, or divide the fractions.

Part 1
Add and subtract fractions with like denominators.

Toss one die. The number showing will be the denominator of the two fractions. Toss both dice. The numbers showing will be the two numerators. Repeat to generate at least five pairs of fractions.

Write an addition number sentence with each pair of fractions and find the sum.

Write a subtraction number sentence with each pair of fractions and find the difference. Subtract the smaller numerator from the larger numerator.

Mathematics for SylvanSync™
Grades 6–8
Fractions with Number Cubes

Part 2

Add and subtract fractions with unlike denominators.

Toss both dice for the first fraction. Choose one number to be the numerator and the other will be the denominator. Toss both dice again to make the second fraction. Repeat to generate at least five pairs of fractions.

Write an addition number sentence with each pair of fractions and find the sum. Write the answer in simplest form.

Compare each pair of fractions to determine which is larger. Write and solve a subtraction number sentence for each pair, subtracting the smaller fraction from the larger fraction.

Part 3

Multiply and divide fractions.

Toss both dice for the first fraction. Choose one number to be the numerator and the other will be the denominator. Toss both dice again to make the second fraction. Repeat to generate at least five pairs of fractions.

Write a multiplication number sentence for each pair of fractions and find the product. Write the answer in simplest terms.

Write and solve a division number sentence with each pair of fractions. Write the answer in simplest form.

Mathematics for SylvanSync™
Grades 6–8
Probability Activities

Objectives
The student will be able to:

- determine the experimental probability of an event.
- determine the theoretical probability of an event.
- compare experimental probability to theoretical probability.

Materials
- a pair of dice

Background Information
The experimental probability of an event is the observed probability, based on an experiment. The experimental probability of a particular outcome is the ratio of the number of desired outcomes to the total number of observed outcomes. The theoretical probability of an event is the expected probability, based on the mathematical likelihood of each outcome. The theoretical probability of a particular outcome is the ratio of the number of desired outcomes to the total number of possible outcomes.

Activity
Complete the activities that follow on a separate sheet of paper.

Perform experiments with dice and compare the experimental probability to the theoretical probability.

Part 1
1. Roll a pair of dice 10 times and record each roll as a number pair.
2. Draw a tree diagram to show the possible outcomes of rolling two dice.
3. What is the theoretical probability of rolling the same number with both dice?
4. What is the theoretical probability of rolling two even numbers?

Mathematics for SylvanSync™
Grades 6–8
Probability Activities

Part 2

A spinner is divided into four equal sections. The sections are red, blue, green, and yellow.

1. What is the theoretical probability of spinning red?
2. What is the theoretical probability of NOT spinning green?
3. What is the theoretical probability of spinning yellow and then blue?
4. What is the theoretical probability of spinning blue, and then spinning yellow or green?

Part 3

1. Roll a die 50 times and record how many times the die lands with each number showing. Use a chart like the one below to record your trials.

Number on Cube	Number of Outcomes
1	
2	
3	
4	
5	
6	

2. What is the theoretical probability of rolling each number?
3. Compare the theoretical probability of rolling each number to the experimental probability of rolling each number. Explain any differences.

Mathematics for SylvanSync™
Grades 6–8
Geometric Solids

Objectives

The student will be able to:

- classify geometric solids.
- identify the two-dimensional shapes that comprise the faces of a geometric solid.
- identify a geometric solid, given the two-dimensional shapes that comprise its faces.

Materials

- wooden GeoSolids
- paper
- crayons
- scissors
- tape

Background Information

A geometric solid is called a three-dimensional shape because it has dimensions of width, depth, and height. Each face of a solid is a two-dimensional shape, such as a square, a rectangle, or a triangle. A solid has a volume, which is the amount of space it takes up or the amount of liquid it could hold; and it has a surface area, which is the amount of material needed to make its faces.

Mathematics for SylvanSync™
Grades 6–8
Geometric Solids

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

1. Look at the different geometric solids. Sort them into three groups: solids that slide; solids that roll; and solids that slide and roll. What do you notice about the solids in each group? Write your answer in a complete sentence.

2. How else can you classify the solids? Write your answer in a complete sentence.

Part 2

1. Choose a geometric solid that slides, but does not roll. Place it on a piece of paper and trace it. What shape is the face you have drawn? Change the position of the solid and trace another face. Is it the same shape as the first face, or is it a different shape? Name all of the shapes that comprise the faces of the solid.

2. Cut out the shapes you have traced. Attach them together to assemble the solid. You may need to cut out more than one of the same shape.

3. Repeat with another solid.

Part 3

1. A square pyramid has a square base and four triangular faces. Which solid is the square pyramid?

2. A cube has six square faces. Which solid is the cube?

3. A hexagonal prism has two hexagonal faces and six rectangular faces. Which solid is the hexagonal prism?

4. Find three other prisms. What do you think they are called?

Mathematics for SylvanSync™
Grades 6–8
Pre-Algebra Concept Practice

Objectives
The student will be able to:

- use mental math to multiply.
- perform operations with integers.
- solve expressions and equations involving variables.

Materials
- Learning Wrap-Ups for Pre-Algebra

Background Information
Mental math strategies help students perform operations without the use of a calculator. One mental math strategy for multiplying a one-digit number by a two-digit number is to choose two easier numbers to multiply in place of the two-digit number. For example, instead of multiplying 4×55, either find the sum of 4×50 and 4×5 or find the difference between 4×60 and 4×5.

To multiply or divide with integers, perform the operation with the absolute values of the integers and then determine the sign of the product or quotient. If the factors (or the dividend and divisor) have the same sign, the product (or quotient) will be positive. If the signs are different, then the product or quotient is negative.

To add integers with like signs, add the absolute values and keep the same sign. To add integers with unlike signs, subtract the absolute values and keep the sign of the number with the greater absolute value. To subtract integers, add the first integer to the opposite of the second integer.

When solving an equation for a variable, keep the equation balanced by performing the same operations to both sides of the equation. To evaluate an expression for a given value of the variable, substitute the given value for the variable in the expression, and then simplify the expression.

Mathematics for SylvanSync™
Grades 6–8
Pre-Algebra Concept Practice

Activity

To use Learning Wrap-Ups for Pre-Algebra, hold the Wrap-Ups at the top of the key and select the specific board you want to wrap. Hold the string next to the first problem on the left and wrap it over the board to the correct answer on the right. Wrap the string behind the board to the next problem on the left. Continue the process until you have solved all of the problems. When you have finished, secure the string in the latch at the bottom of the key and see if the string matches the lines on the back. If it does, then you have answered the problems correctly.

The parts of this activity can be completed in any order.

Part 1

Take the Mental Math (yellow and blue keys) Wrap-Up. Complete the first, second, and third keys.

Part 2

Take the Addition and Subtraction of Positive and Negative Numbers (green keys) and Multiplication and Division of Positive and Negative Numbers (orange keys) Wrap-Ups. Choose three keys to complete in each Wrap-Up. For these keys, the variable always represents the sum, difference, product, or quotient.

Part 3

Take the working with Solving for the Unknown (yellow and red keys) and Understanding Algebraic Expressions (blue keys) Wrap-Ups. Choose three keys to complete in each Wrap-Up.

Mathematics for SylvanSync™
Grades 6–8
Understanding Percent

Objectives
The student will be able to:

- find the percent equivalent of a unit fraction.
- find the percent equivalent of a fraction.

Materials
- Fraction Tower Cubes
- Percent Tower Cubes

Background Information
There are several ways to represent a number between 0 and 1: as a fraction whose numerator is less than its denominator; as a decimal; and as a percent. Just as a fraction or decimal describes a part of a whole, a percent describes a part of 100, where 100 is the whole.

Activity
Complete the activities that follow on a separate sheet of paper.

Part 1
1. Find the Fraction Tower Cubes that represent each fraction. Then, write each fraction as an equivalent fraction with a denominator of 100.

   a. $\frac{1}{4}$

   b. $\frac{1}{10}$

   c. $\frac{1}{5}$

   d. $\frac{1}{2}$

2. Find the Percent Tower Cubes that match the Fraction Tower Cubes for each fraction in Question 1. What do you notice about the number on the Fraction Tower Cubes and your answers to Question 1? Write your answer in a complete sentence.

Mathematics for SylvanSync™
Grades 6–8
Understanding Percent

Part 2

1. Find the Fraction Tower Cubes that represent each fraction. Then, write each fraction as an equivalent fraction with a denominator of 100.

   a. $\frac{3}{4}$

   b. $\frac{2}{5}$

   c. $\frac{7}{10}$

2. Find the Percent Tower Cubes that match the Fraction Tower Cubes for each fraction in Question 1. What do you notice about the number on the Percent Tower Cubes and your answers to Question 1? Write your answer in a complete sentence.

3. Explain your answers to Question 2 in Parts 1 and 2. Write your answer in a complete sentence.

Mathematics for SylvanSync™
Grades 6–8
Understanding Percent

Part 3

1. Use the Percent Tower Cubes to model each percent in as many different ways as you can.
    a. 50%
    b. 25%
    c. 40%
    d. 33.3%
2. For each model, find the equivalent Fraction Tower Cubes. Represent each percent model as a fraction.